

Understanding Eigenvalues and Eigenvectors (for PCA)

Why do we care about these?

In PCA, the goal is to identify directions where the data varies the most. Eigenvectors represent those directions, and eigenvalues tell us how much variation lies along them.

1. Think of your dataset as a cloud of points

Imagine a scatterplot (e.g., height vs. weight). The cloud of points stretches more in one direction than another. PCA identifies the direction of maximum spread—the first eigenvector.

2. Rubber band analogy

Picture a rubber band around the data cloud:

- Longest stretch → first eigenvector
- Amount of stretch → first eigenvalue

The next longest perpendicular stretch becomes the second eigenvector.

3. What is an eigenvector?

An eigenvector is a special direction in the data where a linear transformation only stretches or shrinks the data—it does not rotate them. In PCA, eigenvectors are the new axes that describe the main patterns of variation.

4. What is an eigenvalue?

An eigenvalue measures how much variance the data have along an eigenvector. Large eigenvalue → important direction. Small eigenvalue → often noise.

5. Everyday example

Height and weight tend to increase together. The strongest combined trend becomes the first eigenvector. Its eigenvalue measures how strong the trend is. The next perpendicular trend becomes the second eigenvector.

6. One-sentence summary

Eigenvectors are directions of variation, and eigenvalues measure how much variation lies in each direction.